

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years building Node.js services on GCP — Pub/Sub, Firestore, Cloud Functions, GCS. Led a zero-downtime migration of 20+ production apps from Parse/MongoDB to Firebase/GCP at MTailor, reporting directly to the CTO. Healthcare-AI proof point: built a WhatsApp Business chatbot integrating OpenAI and Google Vision OCR for prescription intake. Now applying the same systems thinking to healthcare backend services at Groq Health.

CORE COMPETENCIES

GCP (Firestore, Pub/Sub, GCS)

Node.js / TypeScript

Python (data migration, tooling)

Event-Driven Systems

Zero-Downtime Migrations

Healthcare AI Integrations

REST APIs + Microservices

Distributed Remote Delivery

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer (US Remote) — Backend on GCP

- Led end-to-end migration of 20+ production applications from Parse/MongoDB to **Firebase/GCP**, zero downtime, reporting directly to the CTO.
- Built real-time two-way sync between MongoDB and Firestore using **Node.js + Google Pub/Sub** with idempotent consumers and back-pressure handling.
- Migrated file storage from S3 to **GCS via a Python script**, preserving referential integrity across millions of objects.
- Shipped a 3D visualisation feature (video overlay + ffmpeg) that increased buyer conversion.
- Saved the company \$5,000/month by migrating all services off AWS to GCP.

MTailor (Contractor)

Mar 2022 – Jul 2022

Contractor — Parse/MongoDB → Firestore Migration

- Built a Python/Node migration script reading MongoDB and writing to Firestore with complex transformation logic.
- Authored Firebase SDK documentation (Node.js + Python) for the engineering team.
- Trained ops team on the new Firebase dashboard and backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Singapore Remote)

- Built ad-hoc jobs querying over 2 million Firestore records to keep customer prices current from Morningstar APIs.
- Extended backend APIs using Firebase Cloud Functions and Express.js.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead — Legal Tech for NGOs

- Led a team of 4 developers building case management + paralegal database systems.
- Architected a microservices backend with **Apache Kafka, Docker, Kubernetes** for FIDA Uganda.
- Built a cross-platform React Native app with Firebase push notifications.

PROJECTS

Healthcare WhatsApp Chatbot — Prescription Intake

Integrated OpenAI for intent parsing + Google Vision OCR for prescription image extraction, exposed via WhatsApp Business API. Node.js service on GCP.

Node.js · OpenAI · Google Vision · WhatsApp Business API · GCP

Real-time Pub/Sub Sync Pipeline (MTailor)

Live MongoDB ↔ Firestore sync supporting parallel production traffic during a 20+ app migration. Idempotent Pub/Sub consumers, schema translation.

Node.js · GCP Pub/Sub · Firestore · MongoDB

FIDA Uganda Case Management — Microservices Backend

Architected a Kafka + Docker + Kubernetes microservices backend for NGO case management across multiple actors.

Node.js · Kafka · Docker · Kubernetes · gRPC

EDUCATION

B.Sc. Computer Engineering — Makerere University

Aug 2018 – Sep 2022

SKILLS

Proficient: Node.js, TypeScript, GCP (Firestore, Pub/Sub, GCS, Cloud Functions), Firebase, Python, MongoDB, SQL, React, Git, Linux

Intermediate: Go, AWS, Rust, React Native **Patterns:** Event-driven systems, zero-downtime migrations, microservices, REST APIs

Languages: English · Kiswahili